GIS-Based Flood Risk and Vulnerability Mapping

Dhaka Metropolitan Region, Bangladesh

Undergraduate Research Project

Department of Urban and Regional Planning

Bangladesh University of Engineering and Technology (BUET)

Framework: UNDRR Risk = Hazard × Exposure × Vulnerability

Data: NASA POWER MERRA-2 | SRTM 30m DEM | ESA WorldCover 2021 | WorldPop 2020 | GADM v4.1

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Abstract

Dhaka, one of the world's most densely populated megacities, faces escalating flood risk driven by low-lying floodplain topography, intense monsoon rainfall, rapid urbanisation, and growing socioeconomic vulnerability. This study develops a comprehensive GISbased flood risk and vulnerability assessment for the Dhaka Metropolitan Region using the UNDRR Risk = Hazard × Exposure × Vulnerability framework across three analytical phases.

Phase 1 analyses 20 years (2004–2023) of NASA POWER MERRA-2 monthly rainfall data at latitude 23.81°N, longitude 90.41°E. Mean annual rainfall is 2,263 mm (CV = 31.8%), with the monsoon season (JJAS) contributing 63% of annual total. The year 2017 was an extreme outlier at 4,631 mm. Mann-Kendall analysis identifies a statistically significant increasing trend in May rainfall (+14.58 mm/yr, $p = 0.008$) and December rainfall (+2.79 mm/yr, $p = 0.019$). Gumbel extreme-value analysis yields return-period estimates of 705 mm (10-year), 815 mm (25-year), and 897 mm (50-year).

Phase 2 constructs a Flood Hazard Index (FHI) from SRTM 30m DEM and ESA WorldCover 2021: Topographic Wetness Index (weight 0.35), elevation (0.30), slope (0.20), and SCS Curve Number (0.15). Results show that 67% of the 960 km² study area falls in the High or Very High hazard class.

Phase 3 integrates WorldPop 2020 exposure and a composite vulnerability index to produce a normalised risk index. Two-tier quantile classification shows that 84.9% of Dhaka's 12.5 million residents live in Very High risk areas. Ward No-63 records the highest mean risk score (0.9397); Dhania ward has the largest population at risk (114,844 people). All top-10 risk wards fall in the High hazard class, indicating that risk is driven by the coincidence of extreme exposure and vulnerability with already hazardous terrain.

Keywords: *flood risk mapping, GIS, vulnerability assessment, Dhaka, UNDRR, Topographic*
Wetness Index, SCS Curve Number, WorldPop, Gumbel distribution, return periods

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Chapter 1: Introduction

1.1 Background and Context

Dhaka is one of the fastest-growing megacities in the world, with a population exceeding 12 million within the metropolitan area. Situated on the floodplain of the Buriganga, Turag, Balu, and Shitalakshya rivers, the city's geography creates an inherent susceptibility to flooding. Rapid and largely unplanned urbanisation has exacerbated this susceptibility by converting natural wetlands and retention areas into impervious built-up surfaces, dramatically increasing surface runoff and reducing the city's natural drainage capacity.

Flooding is Dhaka's most recurrent and economically damaging natural hazard. The 2004 and 2007 monsoon floods inundated large portions of the city, displacing millions and causing extensive damage to infrastructure, housing, and livelihoods. Climate projections consistently suggest that both the frequency and intensity of extreme rainfall events are likely to increase under future warming scenarios, compounding existing risk.

Despite this well-recognised vulnerability, spatially explicit and methodologically rigorous flood risk assessments at the intra-urban ward level remain limited. Most existing studies either focus on regional-scale river flood modelling or lack integration of exposure and vulnerability dimensions as required by internationally recognised risk frameworks.

1.2 Research Objectives

This study addresses the following four objectives:

1. Characterise the spatial and temporal patterns of rainfall in Dhaka (2004–2023) and estimate return-period events for flood scenario design.

2. Map flood hazard across the Dhaka Metropolitan Region using terrain analysis and land-cover derived runoff potential.

3. Assess flood exposure and social vulnerability at the ward level, integrating population density, built-up intensity, and low-elevation exposure.

4. Produce a composite Flood Risk Index following the UNDRR framework and rank wards by composite risk score.

1.3 Study Area and Scope

The study covers the Dhaka Metropolitan Region (EPSG:32646, UTM Zone 46N), encompassing 242 GADM Level 4 administrative wards across approximately 1,154 km² total area, of which approximately 960 km² is land area (permanent water bodies comprising the remaining ~194 km²). All spatial analyses use UTM 46N projection. The temporal scope of rainfall analysis is 2004– 2023 (20 years).

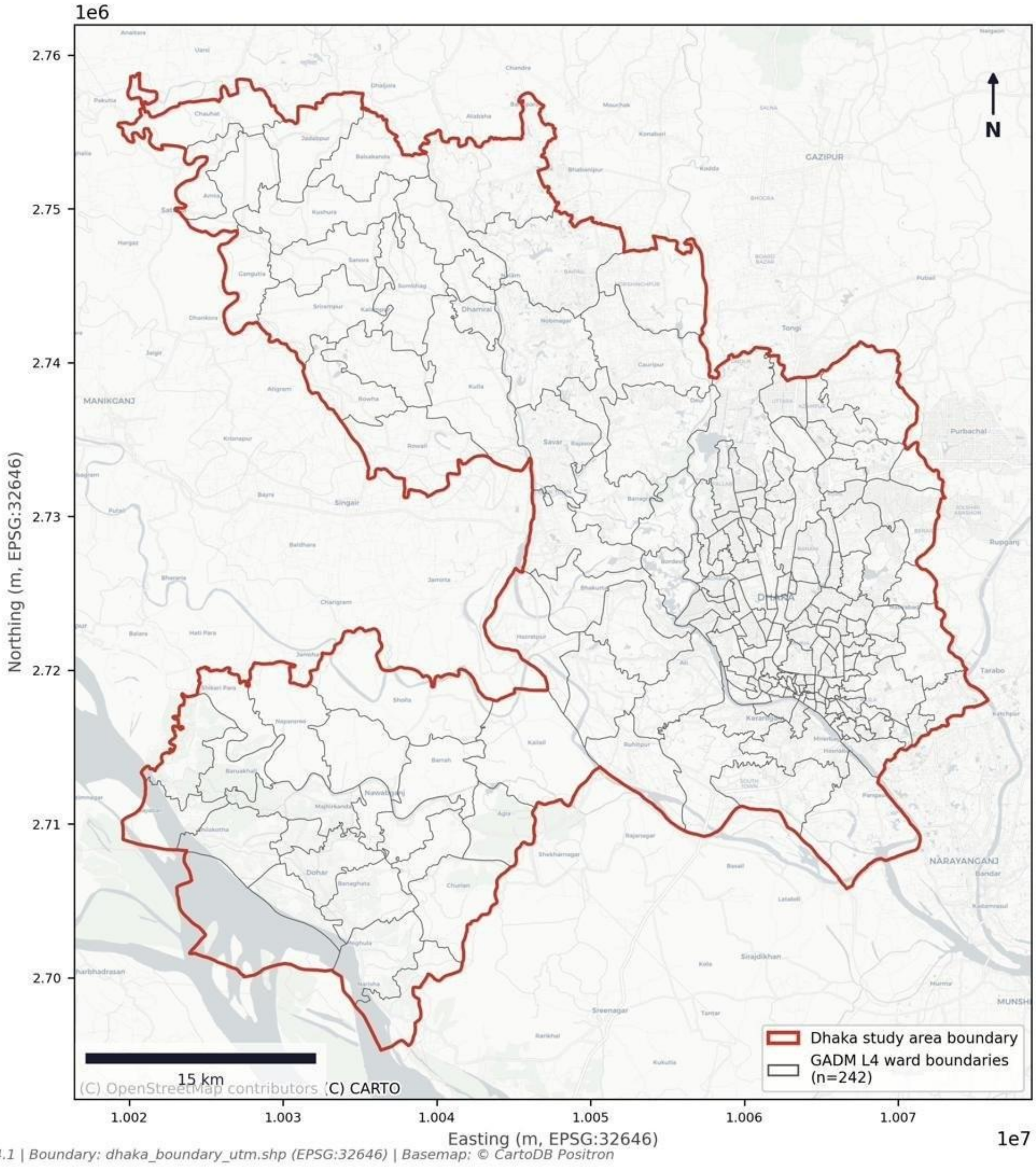
1.4 Report Structure

Chapter 2 describes the study area and data sources. Chapter 3 presents the full methodology. Chapters 4, 5, and 6 present results of the rainfall analysis, flood hazard mapping, and integrated risk assessment. Chapter 7 provides discussion, limitations, and conclusions. References are listed at the end.

Chapter 2: Study Area and Data Sources

2.1 Dhaka Metropolitan Region

The Dhaka Metropolitan Region occupies the central part of Bangladesh within the Bengal Delta at approximately 23.6°–24.0°N and 90.2°–90.6°E. The region is characterised by extremely low-lying floodplain topography, with mean elevation of approximately 8.98 m above mean sea level and over 64% of the area having slopes below 1°. The study area encompasses 242 GADM Level 4 administrative wards. Map 1 shows the study area and ward boundaries.



Data: GADM v4.1 | Boundary: dhaka_boundary_utm.shp (EPSG:32646) | Basemap: © CartoDB Positron

Map 1: Study Area — Dhaka Metropolitan Region. GADM Level 4 ward boundaries (n = 242, grey lines). Red outline = study area boundary. Basemap: © CartoDB Positron. CRS: EPSG:32646 (UTM Zone 46N). Data: GADM v4.1.

2.2 Data Sources

Table 1: *Data sources and specifications used in this study.*

Dataset	Source	Resolution / Period	Use in study
SRTM 30m DEM	NASA/USGS (Earth Explorer)	30m spatial, 2000	Elevation, slope, TWI
ESA WorldCover 2021	ESA Copernicus	10m spatial, 2021	LULC, CN mapping
WorldPop 2020	WorldPop Project, Univ. Southampton	~92m resampled, 2020	Population exposure
NASA POWER MERRA-2	NASA Langley Research Center	0.5°×0.625°, 2004– 2023	Monthly rainfall
GADM v4.1 (Level 4)	GADM.org	Ward polygons, 2023	Zonal statistics
Study boundary (UTM)	Project derived	EPSG:32646 polygon	Clipping / geoprocessing

2.3 Coordinate Reference Systems

All data storage uses EPSG:4326. All geoprocessing uses EPSG:32646 (UTM Zone 46N). Rasters are aligned to the dem_clipped.tif reference grid (30m, EPSG:32646) before multilayer analysis. SRTM nodata = −32,767. WorldPop is resampled using sum aggregation; ESA WorldCover uses nearest-neighbour resampling.

Chapter 3: Methodology

3.1 Analytical Framework

The study adopts the UNDRR conceptual framework:

Risk = Hazard × Exposure × Vulnerability

Each component is normalised to 0–1 before multiplication. The final risk index is normalised to 0–1 and classified using a two-tier quantile scheme to account for the extreme right-skew of multiplicative risk indices.

3.2 Phase 1: Rainfall Analysis

3.2.1 Data and Descriptive Statistics

Monthly cumulative rainfall was obtained from NASA POWER MERRA-2 (23.81°N, 90.41°E) for January 2004 to December 2023 (240 observations). Annual totals were computed by summing 12 monthly values. Descriptive statistics (mean, SD, CV, min, max) were calculated at monthly and annual scales.

3.2.2 Trend Analysis

The Mann-Kendall non-parametric test was applied to each calendar month's time series and to the annual series. Sen's slope estimator quantified the rate of change in mm/year.

3.2.3 Return Period Analysis

Annual peak monthly rainfall was extracted as the extreme-value series. The Gumbel (Extreme Value Type I) distribution was fitted using maximum likelihood estimation and assessed with the Kolmogorov-Smirnov test (KS $p = 0.99$). Return-period quantiles were estimated for 2-, 5-, 10-, 25-, 50-, and 100-year events.

3.3 Phase 2: Flood Hazard Mapping

3.3.1 DEM Preprocessing

The SRTM 30m DEM was reprojected to EPSG:32646 and clipped to the study boundary in QGIS. Hydrological sink filling used GRASS GIS r.fill.dir. The sink-filled dem_clipped.tif is the reference grid for all subsequent rasters.

3.3.2 TWI and Slope

TWI was calculated using GRASS GIS r.watershed with the multiple-flow-direction algorithm. $TWI = \ln(\alpha / \tan \beta)$, where α is the specific catchment area and β is local slope angle.

3.3.3 SCS Curve Number

ESA WorldCover 2021 (10m) was resampled to the DEM grid with nearest-neighbour interpolation. CN values were assigned per class for Hydrologic Soil Group B: Built-up = 85, Cropland = 72, Grassland = 61, Shrubland = 65, Tree cover = 55, Bare/sparse = 77. Permanent water bodies were excluded from FHI computation.

3.3.4 Flood Hazard Index

All four rasters were min-max normalised to 0–1. Elevation and slope were inverted so lower values yield higher hazard. The FHI formula:

$$FHI = 0.35 \cdot TWI\_norm + 0.30 \cdot (1 - Elev\_norm) + 0.20 \cdot (1 - Slope\_norm) + 0.15 \cdot CN\_norm$$

FHI was classified into four equal-interval classes: Low (<0.25), Medium (0.25–0.50), High (0.50–0.75), Very High (>0.75).

3.4 Phase 3: Risk Assessment

3.4.1 Exposure Index

WorldPop 2020 was aligned to the DEM grid using sum resampling. Permanent water bodies were masked and assigned zero exposure. The exposure index is the min-max normalised population density per pixel.

3.4.2 Vulnerability Index

Three normalised sub-indicators: population density (weight 0.40), built-up intensity from ESA WorldCover class 50 (weight 0.35), and low-elevation exposure for pixels ≤4 m (weight 0.25).

3.4.3 Risk Index and Classification

Risk = haz_norm × exp_norm × vuln_norm, normalised to 0–1. Two-tier quantile classification: zero pixels = No Risk; non-zero pixels split at 25th, 50th, 75th percentiles into Low, Medium, High, Very High. Zonal statistics computed for 225 of 242 wards (17 excluded: NULL or insufficient-pixel geometries).

Chapter 4: Phase 1 Results: Rainfall Analysis

4.1 Annual Rainfall Patterns

Mean annual rainfall (2004–2023) is 2,263 mm (SD = 719 mm, CV = 31.8%). The year 2017 was an extreme outlier at 4,631 mm — nearly double the long-term mean — driven by exceptional rainfall in all four monsoon months. The driest year was 2012 (1,507 mm). Figure 1 shows the annual time series.

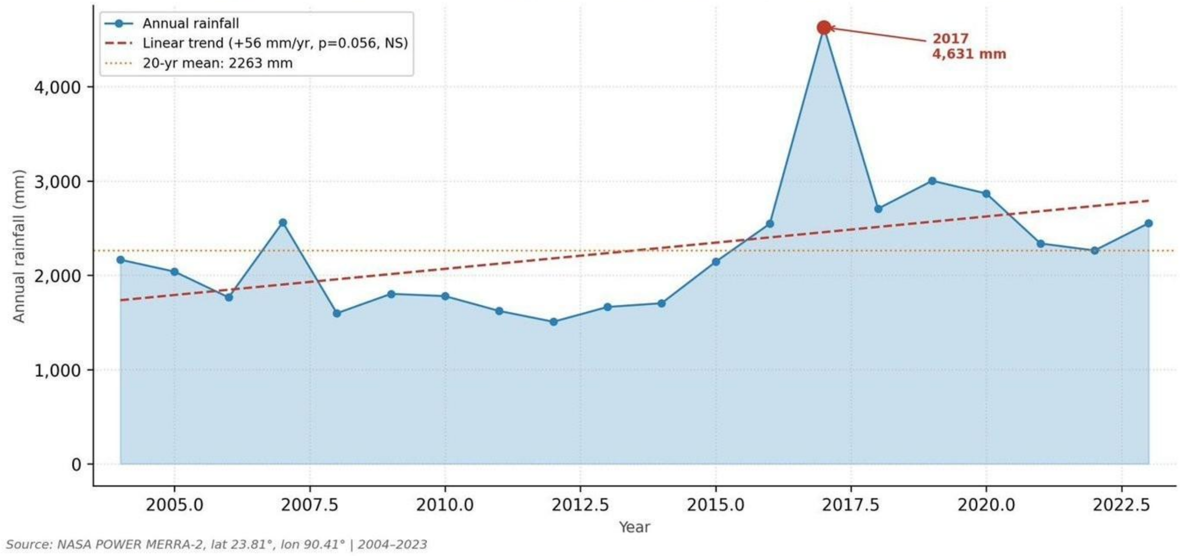


Figure 1: Annual rainfall totals, Dhaka 2004–2023 (NASA POWER MERRA-2, lat 23.81°, lon 90.41°). Dashed red line = linear trend (+56 mm/yr, p = 0.056, not significant). Orange dotted line = 20-year mean (2,263 mm). Red point = 2017 outlier (4,631 mm).

4.2 Seasonal Distribution

The monsoon season (JJAS) contributes 63% of mean annual rainfall. July is the single wettest month (mean 405 mm), followed by June (386 mm) and August (354 mm). The dry season (November–March) accounts for less than 5% of annual rainfall. Figure 2 shows monthly distributions across all 20 years.

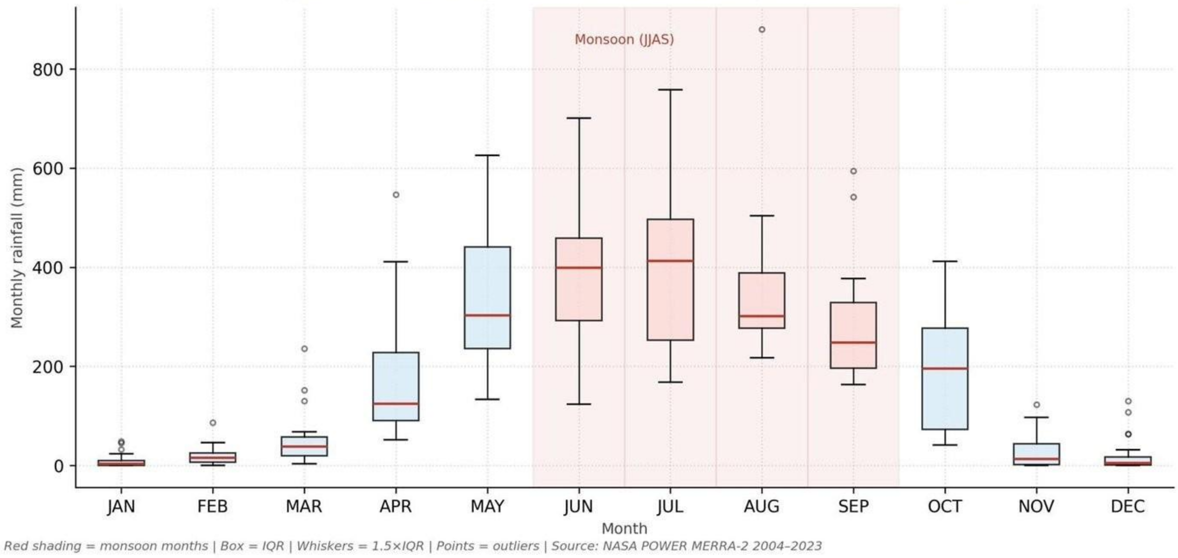


Figure 2: Monthly rainfall distribution, Dhaka 2004–2023. Red-shaded months = monsoon (JJAS). Box = IQR; whiskers = 1.5×IQR; points = outliers. Source: NASA POWER MERRA-2.

4.3 Trend Analysis

The Mann-Kendall test on the annual series yields Z = 1.91, p = 0.056 (not significant). Monthly analysis identifies two statistically significant trends: May rainfall increasing at +14.58 mm/yr (Z = 2.66, p = 0.008), and December rainfall increasing at +2.79 mm/yr (Z = 2.34, p = 0.019). The May trend is particularly significant — intensifying pre-monsoon rainfall saturates soils before the main monsoon onset, reducing the system's capacity to absorb subsequent extreme events. Table 2 presents the full monthly statistics.

Table 2: Monthly rainfall descriptive statistics and Mann-Kendall trend results, Dhaka 2004–2023.

Month	Mean (mm)	SD (mm)	CV (%)	Min (mm)	Max (mm)	MK Z	p-value	Slope (mm/yr)	Sig.
JAN	9.9	15.3	155.0	0.0	48.8	1.62	0.105	0.78	No
FEB	19.9	20.7	104.3	0.0	86.8	0.71	0.475	0.88	No
MAR	52.8	57.3	108.5	3.3	235.8	1.49	0.136	1.95	No
APR	173.2	125.6	72.5	52.2	546.7	0.45	0.650	3.57	No
MAY	329.8	134.2	40.7	133.2	625.7	2.66	0.008	14.58	Yes*
JUN	386.1	132.7	34.4	124.0	700.7	1.95	0.052	10.56	No

JUL	404.6	186.1	46.0	168.2	758.2	0.13	0.897	3.82	No
AUG	353.5	147.2	41.6	217.0	880.2	1.95	0.052	8.16	No
SEP	286.4	119.2	41.6	163.1	594.3	0.84	0.399	0.19	No
OCT	195.7	115.3	58.9	41.3	411.5	1.04	0.299	6.15	No
NOV	28.4	37.1	130.5	0.0	122.7	1.49	0.136	2.10	No
DEC	22.3	38.3	171.5	0.0	129.8	2.34	0.019	2.79	Yes*

* Significant at $p < 0.05$.

4.4 Return Period Analysis

The Gumbel distribution was fitted to the 20-year peak monthly maxima series (KS p = 0.99, excellent fit). Figure 3 shows the fitted curve with observed Gringorten plotting positions and design events. Table 3 presents the full return period estimates.

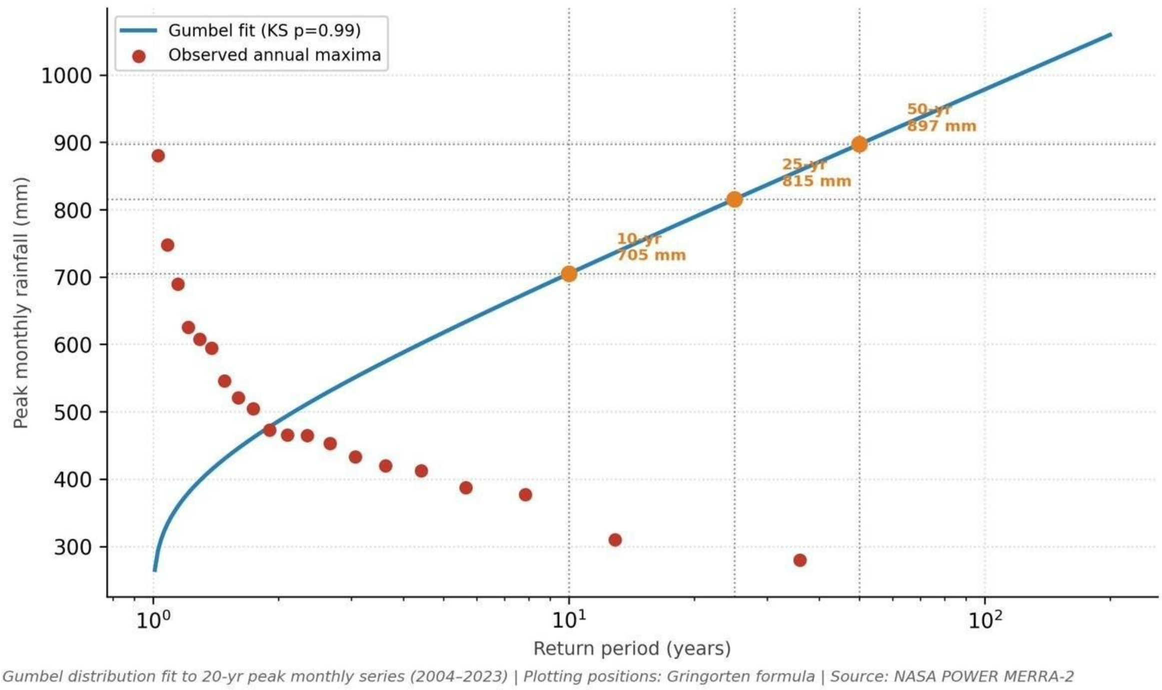


Figure 3: Gumbel return period curve for Dhaka peak monthly rainfall. Blue line = fitted Gumbel (KS p = 0.99). Red points = observed annual maxima (Gringorten plotting positions). Orange markers = 10-, 25-, and 50-year design events. Source: NASA POWER MERRA-2 2004–2023.

Table 3: Gumbel return period estimates for peak monthly rainfall, Dhaka.

Return period (yr)	Exceedance probability	Peak monthly rainfall (mm)	Application
2	1/2 (50%)	486	Frequent event — baseline scenario
5	1/5 (20%)	618	Moderate event
10	1/10 (10%)	705	Phase 2 low design scenario
25	1/25 (4%)	815	Phase 2 medium design scenario
50	1/50 (2%)	897	Phase 2 high design scenario
100	1/100 (1%)	978	Extreme event — planning horizon

Chapter 5: Phase 2 Results: Flood Hazard Mapping

5.1 Topographic Characteristics

The Dhaka Metropolitan Region is characterised by extremely flat, low-lying terrain. Mean elevation is 8.98 m above mean sea level. Over 64% of the study area has slopes below 1°. The Topographic Wetness Index ranges from 2.8 to 20.3 with a mean of 8.32. The highest TWI values are concentrated along river banks and in the eastern chars — areas most prone to water accumulation during rainfall events.

5.2 FHI Component Weights

Table 4: Flood Hazard Index component weights and physical rationale.

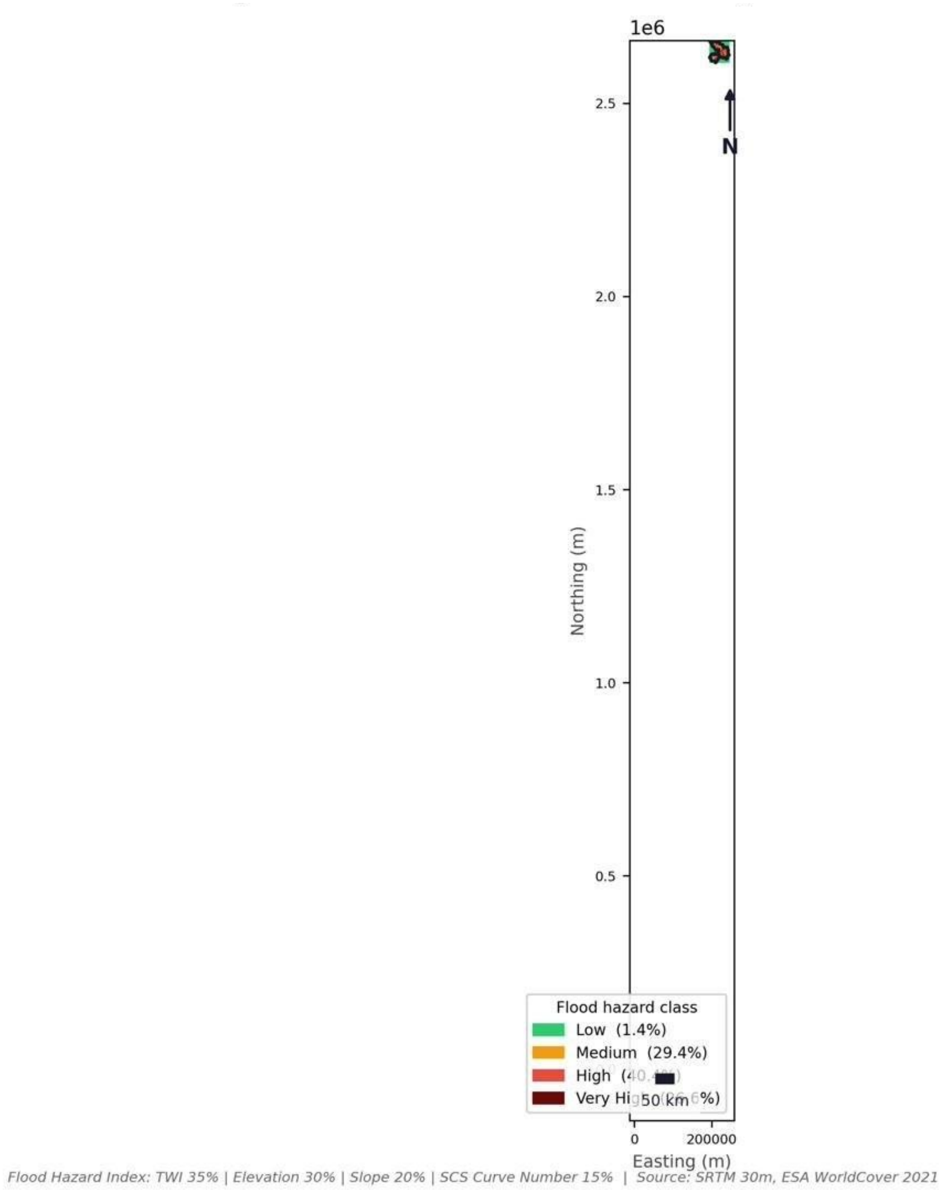
Component	Weight	Physical rationale
Topographic Wetness Index (TWI)	0.35 (35%)	Primary indicator of water accumulation potential in flat deltaic terrain
Elevation	0.30 (30%)	Low elevation directly increases susceptibility to prolonged inundation
Slope	0.20 (20%)	Flat slopes reduce drainage velocity and prolong surface water retention
SCS Curve Number (CN)	0.15 (15%)	Land-cover effect on infiltration capacity and direct runoff generation

5.3 Hazard Classification Results

The composite FHI shows that 67% of the study area falls in the High or Very High hazard class. The Very High class (26.6%, ~256 km²) is concentrated in the eastern and northeastern parts of the city, corresponding to the Balu River floodplain and historically floodprone chars. Table 5 presents class areas and Map 3 shows the spatial distribution.

Table 5: Flood hazard class areas and FHI value ranges.

Hazard class	Class value	Area (km²)	% of study area	FHI range
Low	1	~14	1.4%	< 0.25
Medium	2	~282	29.4%	0.25 – 0.50
High	3	~388	40.4%	0.50 – 0.75
Very High	4	~256	26.6%	> 0.75
Total	—	~960	100%	—

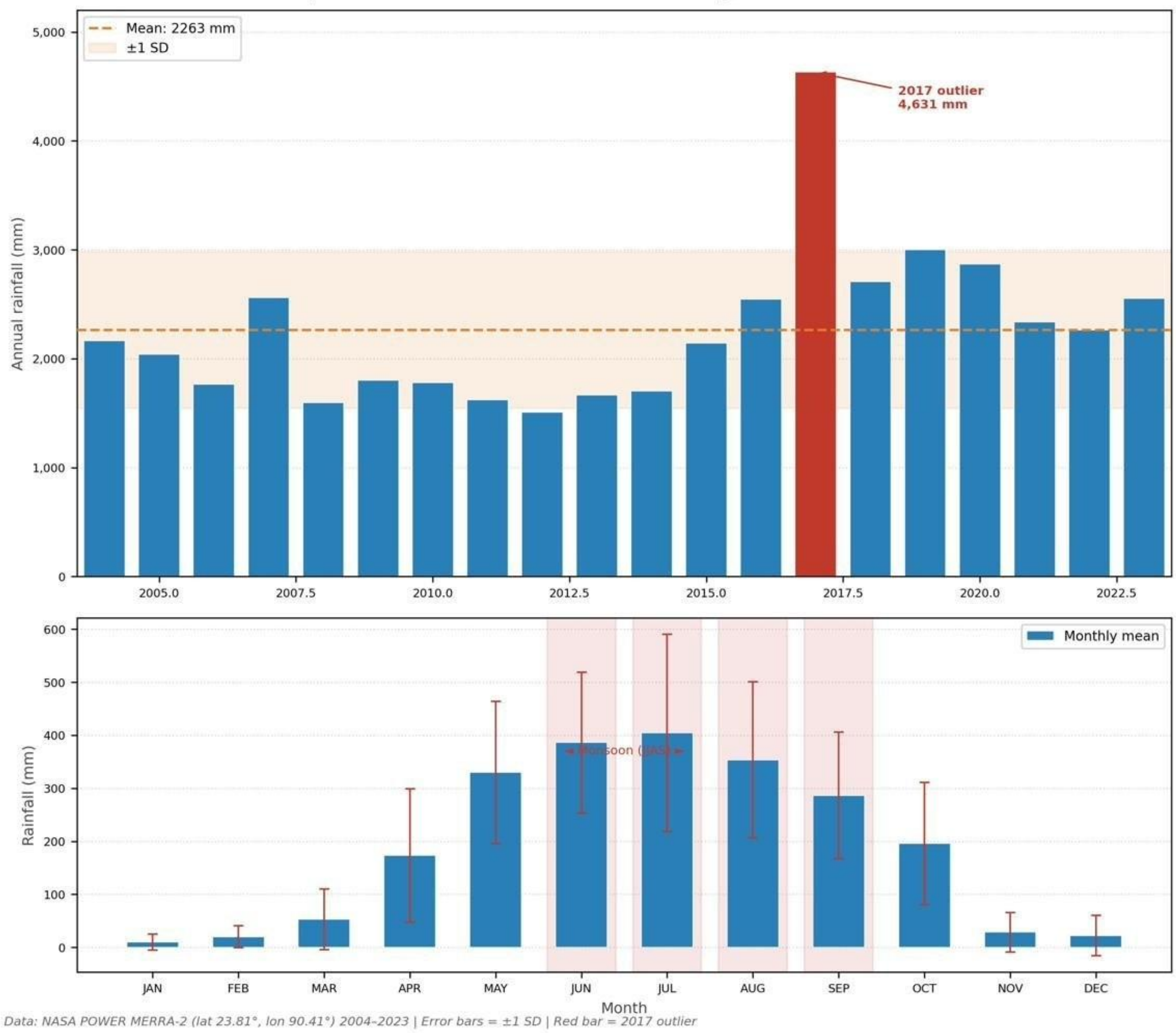


Map 2: Flood Hazard Classification, Dhaka Metropolitan Region. FHI = TWI (35%) + Elevation (30%) + Slope (20%) + SCS Curve Number (15%). Source: SRTM 30m DEM, ESA WorldCover 2021.

Chapter 6: Phase 3 Results: Exposure, Vulnerability and Flood Risk

6.1 Rainfall Variability Map

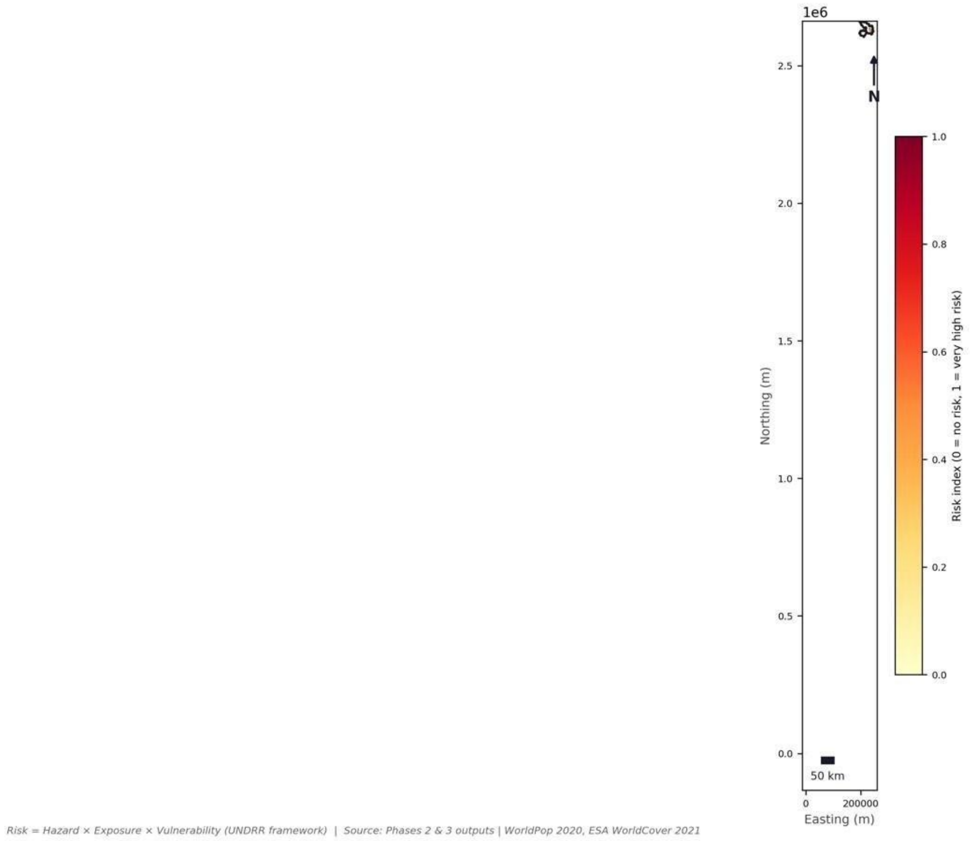
Map 2 summarises the annual and seasonal rainfall patterns that drive the hazard context. The upper panel shows annual totals with the 2017 outlier; the lower panel shows mean monthly totals with monsoon months highlighted.



Map 3: Annual Rainfall Variability, Dhaka 2004–2023. Upper: annual totals with mean and ±1 SD band; 2017 outlier annotated. Lower: mean monthly totals with ±1 SD error bars; monsoon months (JJAS) shaded. Source: NASA POWER MERRA-2.

6.2 Composite Risk Index

The composite risk index (Risk = Hazard × Exposure × Vulnerability, normalised 0–1) is shown as a continuous surface in Map 4. The index is strongly right-skewed, with most non-zero pixels between 0.4 and 1.0, reflecting the co-occurrence of high hazard, dense population, and extensive built-up coverage across Dhaka's residential fabric.



Map 4: Composite Flood Risk Index, Dhaka Metropolitan Region. Continuous scale 0 (no risk) to 1 (very high risk). Risk = Hazard × Exposure × Vulnerability (UNDRR framework). Source: Phase 2 and Phase 3 outputs, WorldPop 2020, ESA WorldCover 2021.

6.3 Risk Class Breakdown

The two-tier quantile classification assigns zero pixels to No Risk and splits non-zero pixels at the 25th, 50th, and 75th percentiles. The key finding is that 84.9% of the 12.5 million residents fall in the Very High risk class at a mean density of 39,481 people/km². Table 6 presents the full breakdown.

Table 6: Flood risk class breakdown by area, population, and population density.

Risk class	Area (km²)	Population	% of total population	Population density (ppl/km²)
No Risk	78	46,896	0.4%	601
Low	269	269,519	2.2%	1,002
Medium	269	497,344	4.0%	1,849
High	269	1,074,155	8.6%	3,993
Very High	269	10,620,356	84.9%	39,481
Total	1,154	12,508,270	100%	10,839

Note: Total study area in Table 6 (1,154 km²) includes permanent water bodies (~194 km²) classified as No Risk due to zero population exposure. Phase 2 hazard mapping covers land area only (~960 km²), from which water pixels were excluded before FHI calculation. Both figures are correct and refer to different analytical extents.

The Hazard × Risk cross-tabulation (Figure 4) reveals that even within the Medium hazard class, substantial populations are classified as Very High risk — because exposure and vulnerability are so extreme that they elevate risk across all hazard zones.

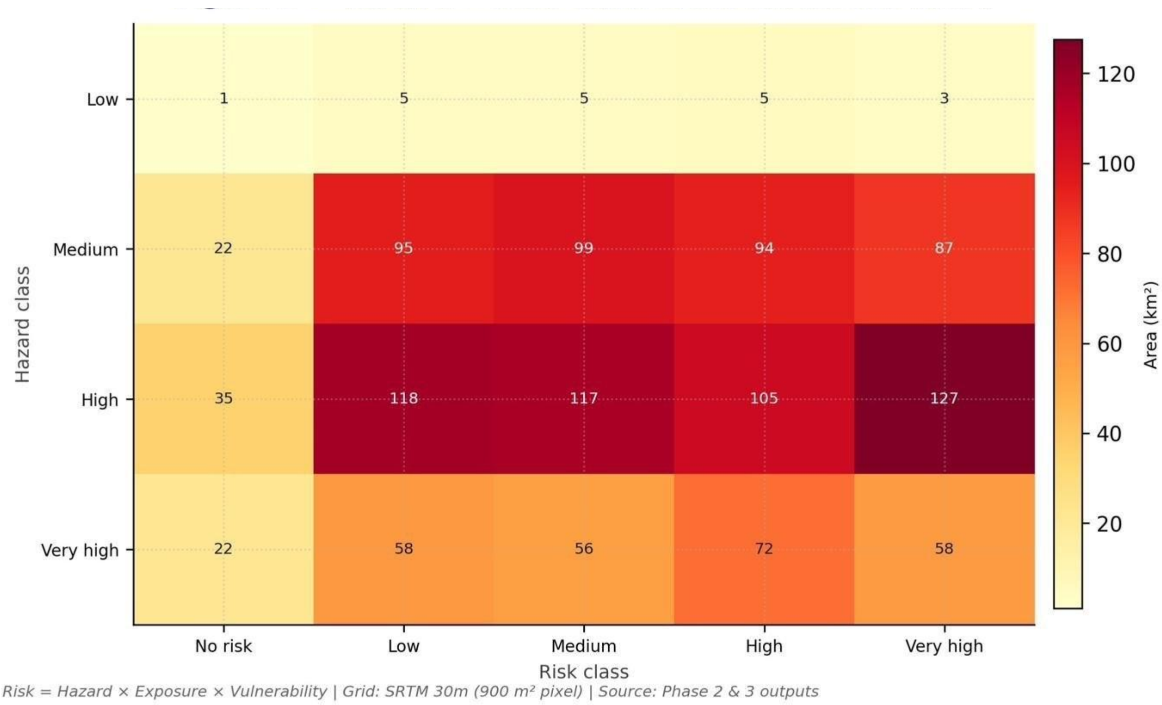
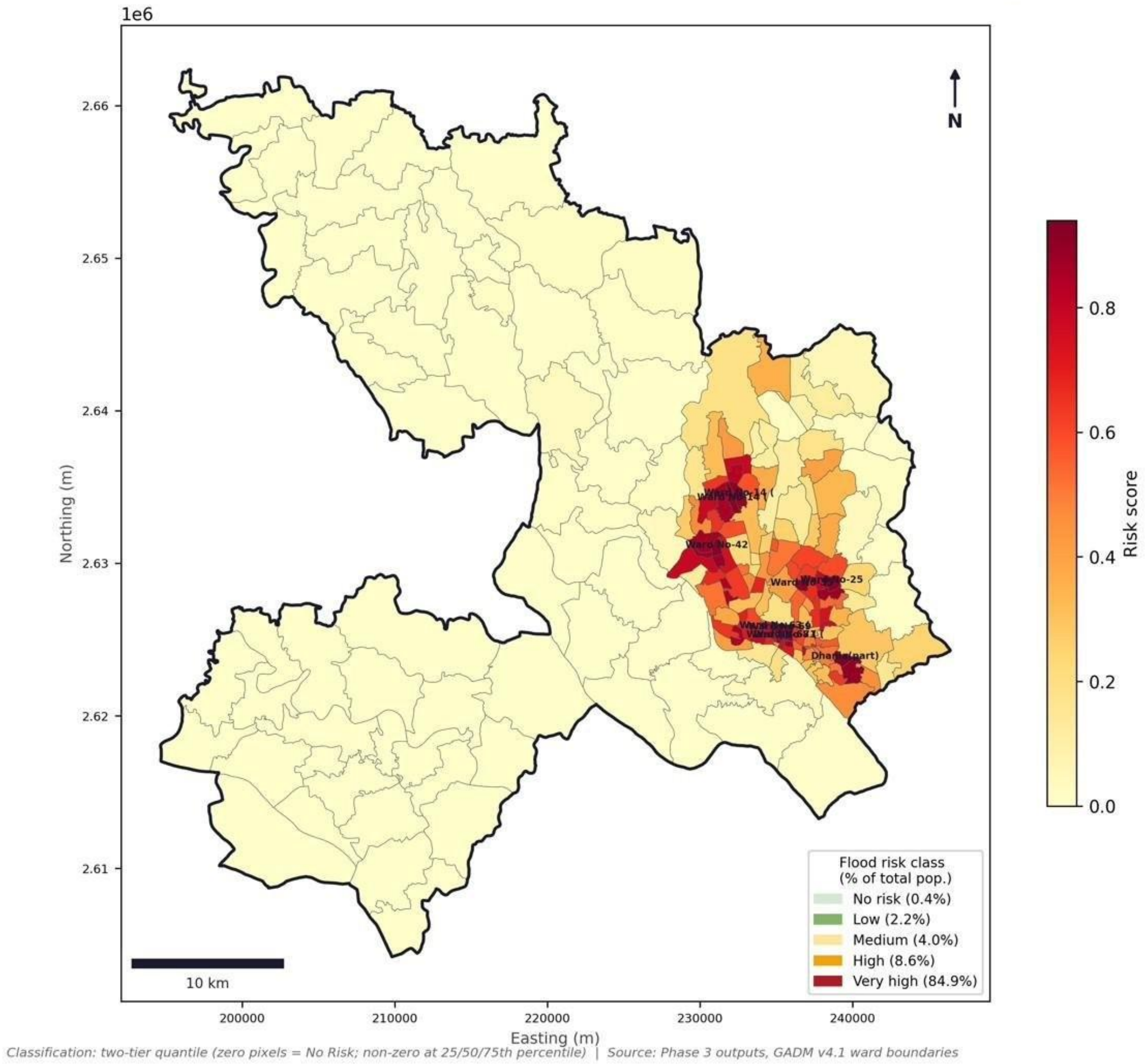


Figure 4: Hazard × Risk Class Cross-Tabulation (km²). Values show area falling in each hazard–risk combination. Grid: SRTM 30m (900 m² per pixel). Source: Phase 2 and Phase 3 outputs.

6.4 Ward-Level Risk

Map 5 shows the ward-level risk classification. The highest concentrations are in eastern wards centred on Demra and Jatrabari, combining High hazard terrain with extreme population density and extensive informal built-up coverage.



Map 5: Ward-Level Flood Risk Classification, Dhaka Metropolitan Region. Two-tier quantile classification (zero pixels = No Risk; non-zero at 25/50/75th percentile). Legend shows percentage of total population. Source: Phase 3 outputs, GADM v4.1.

Figure 5 ranks the top 10 wards by mean risk score and population at risk. Ward No-63 (Part) records the highest score (0.9397) but small population (16,039) as it retains only the highest-risk pixels after boundary clipping. Dhania (Part) has both a very high score (0.9323) and the largest population at risk (114,844 people). All 10 wards fall in the High hazard class, confirming that maximum composite risk is driven by extreme exposure and vulnerability, not by the zones of maximum hazard.

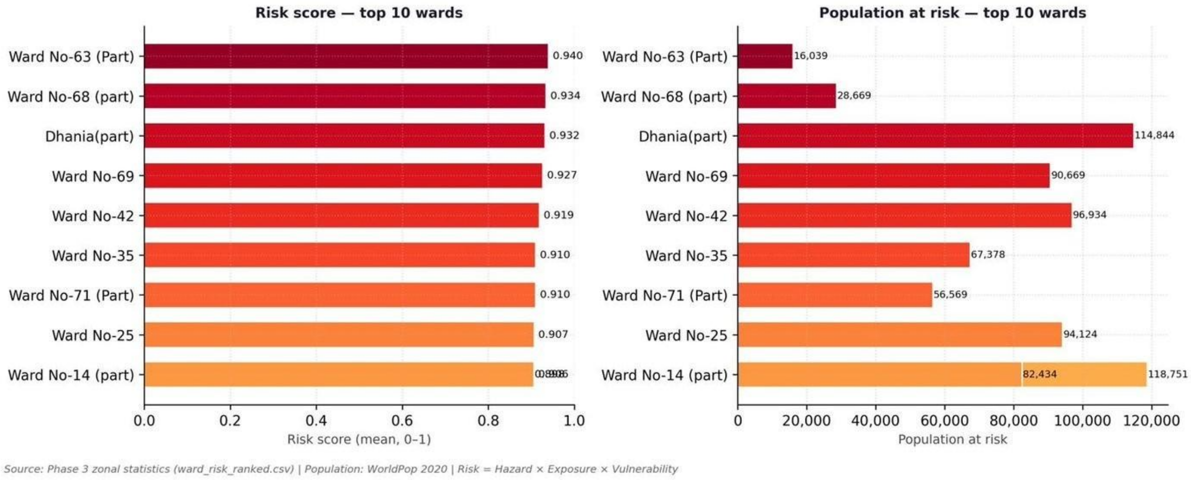


Figure 5: Top 10 wards by mean flood risk score (left) and total population at risk (right). Note: Ward No-14 appears twice due to two partial polygon fragments after study boundary clipping. Source: Phase 3 zonal statistics, WorldPop 2020.

Table 7: Top-10 wards by mean flood risk score with supporting metrics.

Rank	Ward	Risk (mean)	Risk (max)	Population	Hazard class	Vulnerability	Exposure
1	Ward No-63 (Part)	0.9397	1.0	16,039	High	0.9987	0.9975
2	Ward No-68 (part)	0.9342	1.0	28,669	High	0.9973	1.0000
3	Dhania (part)	0.9323	1.0	114,844	High	0.9669	0.9480
4	Ward No-69	0.9266	1.0	90,669	High	0.9978	1.0000
5	Ward No-42	0.9189	1.0	96,934	High	0.9853	1.0000
6	Ward No-35	0.9104	1.0	67,378	High	0.9849	0.9932
7	Ward No-71 (Part)	0.9098	1.0	56,569	High	0.9917	1.0000
8	Ward No-25	0.9066	1.0	94,124	High	0.9616	0.9310
9	Ward No-14 (part)	0.9063	1.0	82,434	High	0.9768	0.9566

10	Ward No-14 (part)	0.8977	1.0	118,751	High	0.9707	0.9497

* Rows 9 and 10 both show Ward No-14 because two partial polygon fragments were retained after clipping. The larger fragment (118,751 people) is the policy-relevant unit.

Chapter 7: Discussion, Limitations and Conclusions

7.1 Synthesis of Findings

The three-phase analysis consistently points to eastern and north-eastern Dhaka as the zone of greatest flood risk. This pattern reflects the convergence of: (1) low-lying floodplain topography with high TWI and minimal slope; (2) extreme population densities driven by rapid in-migration and informal settlement growth; and (3) extensive impervious built-up coverage that suppresses infiltration and amplifies direct runoff.

The finding that 84.9% of Dhaka's population lives in the Very High risk class is not an artefact of classification. The two-tier quantile scheme was designed to prevent inflation of higher classes, yet still produces this result — consistent with the physical and demographic reality of one of the world's most flood-exposed megacities.

The significant increasing trend in May pre-monsoon rainfall (+14.58 mm/yr) represents a meaningful shift in the seasonal flood risk calendar. Pre-monsoon rainfall saturates soils before JJAS onset, reducing system capacity to absorb subsequent extreme events. Flood early warning systems in Dhaka should begin heightened monitoring protocols from early May, not June.

The Gumbel analysis establishes that even a 10-year design event (705 mm peak monthly) represents extreme stress on Dhaka's drainage infrastructure. Urban drainage design standards in Bangladesh are typically calibrated for 2–5 year return periods, leaving a substantial gap relative to observed hazard.

7.2 Limitations

- DEM resolution: SRTM 30m does not resolve fine-scale urban drainage features (culverts, khals, retention ponds). LiDAR or high-resolution commercial DEM would significantly improve TWI and slope accuracy in near-flat terrain.
- Static population data: WorldPop 2020 is a single-year snapshot. Dhaka's population is growing rapidly in the eastern wards, so current exposure is likely underestimated.
- Single rainfall point: NASA POWER MERRA-2 provides data at one 0.5°×0.625° grid point, missing sub-city spatial variability in convective rainfall.
- No drainage infrastructure layer: The study excludes the existing stormwater network, pumping stations, and WASA retention infrastructure, all of which modify effective flood hazard.
- Short record for extreme value analysis: The 20-year record limits reliability of return-period estimates beyond 50 years.
- Partial ward geometries: Ward No-14 appears twice; several other wards marked (Part) reflect study boundary clipping. Scores represent only pixels within the study area.

7.3 Directions for Future Research

- Sentinel-1 SAR flood inundation data to validate the hazard map against observed historical inundation extents.
- Hydrodynamic modelling (HEC-RAS 2D or MIKE FLOOD) using Phase 1 design storms to produce scenario-specific inundation depth and duration maps.
- Expansion of the vulnerability index to incorporate socioeconomic indicators (income, education, housing quality) from Bangladesh Census data.
- LULC change analysis (2010–2023) to quantify the contribution of urbanisation to increasing CN and flood hazard over time.
- Climate change scenario analysis using CMIP6 projections under SSP2-4.5 and SSP5-8.5 to estimate future shifts in rainfall regimes and return-period magnitudes.

7.4 Conclusions

This study provides the first comprehensive, ward-level flood risk assessment of the Dhaka Metropolitan Region using the UNDRR framework and freely available global datasets. Five principal conclusions emerge:

1. Dhaka's rainfall regime features intense monsoon concentration (63% in JJAS, CV = 31.8%) and a statistically significant increasing trend in May pre-monsoon rainfall (+14.58 mm/yr, p = 0.008), indicating a lengthening flood risk season. A 10-year return period peak monthly event is estimated at 705 mm.
2. Flood hazard is pervasive: 67% of the study area is in the High or Very High hazard class, driven by the city's flat floodplain setting and high surface runoff potential.
3. Approximately 84.9% of Dhaka's 12.5 million residents live in Very High flood risk areas — a consequence of extreme population density and built-up coverage overlapping with hazardous terrain.
4. The highest composite risk scores are in the eastern and north-eastern wards. Ward No-63 has the highest mean risk score (0.9397); Dhania ward has the largest population at risk (114,844 people). All top-10 wards fall in the High hazard class, confirming that risk is driven by exposure–vulnerability coincidence, not by the extreme hazard zones alone.
5. Risk reduction must address the convergence of physical hazard with extreme demographic and built-environment vulnerability. Structural interventions alone will be insufficient without land-use regulation, building codes, and communitylevel resilience measures in the high-risk eastern wards.

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